
SIGMA0: ONTIC FOUNDATIONS OF CONSTRAINT-SURFACE DYNAMICS

A PREPRINT

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May 24, 2026

ABSTRACT

Σ_0 states the ontic commitments underlying Constraint-Surface Dynamics. The paper treats physical reality as deterministic Hamiltonian evolution on a compact finite-dimensional Kähler manifold Σ , equipped with Liouville measure μ_L , while quantum state space $\mathbb{CP}^{N-1}$ is interpreted as an epistemic projection through a smooth many-to-one map $\pi : \Sigma \rightarrow \mathbb{CP}^{N-1}$. Its purpose is not to derive quantum mechanics, spacetime, or relativistic structure from first principles, but to make explicit the assumptions required by the existing finite-dimensional CSD programme.

The paper defines the roles of Σ , π , μ_L , Σ -time, Σ -energy, measurement, and contextual nonlocality. Measurement is interpreted as a Σ -level de-isolation event, while spacetime measurement records are treated as coarse-grained images of deeper ontic dynamics. Bell-type correlations are therefore accepted as nonlocal relative to emergent spacetime, while leaving open whether they are structurally local in Σ .

The contribution of Σ_0 is deliberately limited. It does not derive Σ , π , spacetime, c , Planck-scale resolution, cosmological-scale bounds, or the mechanism connecting Σ -local structure to spacetime-distant correlations. Instead, it provides an internal ontological foundation for later work on approximate projectability, emergent spacetime, causal order, and effective locality.

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1 Introduction

1.1 Purpose of the Paper

This paper states the ontic commitments underlying Constraint-Surface Dynamics. Its purpose is not to derive the full quantum formalism, spacetime, or relativistic structure from first principles. Its purpose is narrower: to make explicit the physical picture assumed by the existing finite-dimensional CSD programme.

The central posit is that the fundamental physical state is an ontic microstate

$$x \in \Sigma,$$

where Σ is a compact finite-dimensional Kähler manifold equipped with Liouville measure μ_L and deterministic Hamiltonian flow. The quantum state is not treated as the fundamental physical state. Rather, the projective state

$$[\psi] \in \mathbb{CP}^{N-1}$$

is interpreted as an epistemic image of deeper ontic structure under a smooth many-to-one projection

$$\pi : \Sigma \rightarrow \mathbb{CP}^{N-1}.$$

This distinction is the organising principle of the paper. The ontic level contains the actual state and deterministic dynamics. The epistemic level contains the projective quantum description. The operational level contains laboratory probabilities, measurement records, and effective quantum predictions.

1.2 Why Σ_0 Is Needed

The existing CSD corpus already contains several technical layers: deterministic volume typicality, the finite-dimensional measure bridge, Born-form operational weights, mixed states, POVMs, sequential update, Bell-sector constructions, and formal Lean milestones. These results rely on a deeper ontological picture, but that picture should not remain implicit.

Without an explicit ontic foundation, later papers risk two errors. First, they may appear to derive structures that are currently assumed. Second, they may blur the distinction between quantum-effective results and deeper claims about spacetime, locality, and causation.

Σ_0 is therefore an internal clarification document. It separates:

what CSD posits

from

what CSD has derived

and from

what CSD still needs to derive.

This separation is essential because the deepest objects in the programme, especially Σ , π , and the emergence of spacetime, are not yet derived from more primitive principles.

1.3 Core Ontological Commitments

The paper adopts the following commitments.

First, physical reality is represented by deterministic dynamics on Σ . A physical system has an actual ontic state x , and its evolution is generated by Hamiltonian flow on the ontic manifold.

Second, the Liouville measure μ_L is the relevant invariant volume measure on Σ . It supplies the measure-theoretic backbone for volume typicality and frequency convergence.

Third, quantum state space is an epistemic projection of the ontic structure. The map

$$\pi : \Sigma \rightarrow \mathbb{CP}^{N-1}$$

identifies ontic states that are operationally indistinguishable at the projective quantum level.

Fourth, measurement is interpreted as a Σ -level de-isolation event. A subsystem that was approximately projectable ceases to evolve autonomously when it becomes dynamically coupled to apparatus or environmental degrees of freedom.

Fifth, CSD is contextual and nonlocal relative to emergent spacetime. It does not assume Bell-local outcome functions. It instead treats outcomes as context-indexed consequences of the realised system-apparatus interaction.

Sixth, spacetime is treated as emergent rather than primitive. The detailed map from Σ -level structure to spacetime structure is not specified in this paper.

1.4 Main Non-Claims

The status of this paper is deliberately conservative. Σ_0 does not claim to complete the CSD programme.

It does not derive the existence of Σ . It does not derive the Kähler structure of Σ . It does not derive the projection π . It does not derive spacetime, Lorentzian metric structure, or relativistic causal cones. It does not prove that the Σ -level causal rate is identical to the observed speed of light. It does not derive Planck-scale or cosmological-scale resolution bounds. It does not provide a complete Σ -level mechanism for Bell correlations.

These are not treated as hidden results. They are explicitly identified as research targets.

The paper should therefore be read as an ontological foundation and assumptions audit, not as a final first-principles derivation.

1.5 Position Within the CSD Programme

Σ_0 sits below the existing technical reconstruction papers. Paper A supplies the deterministic repeated-trial and volume-typicality layer. Paper B and TN1 address the finite-dimensional measure and Born-weight layer. Paper C gives the reconstruction architecture. Paper D gives the public ontological synthesis. TN4, TN6, and TN7 extend the operational and many-mode finite-dimensional dictionary. LF1, LF2, and LF3 formalise selected theorem layers.

The role of Σ_0 is different. It states what these layers are about physically.

In particular, Σ_0 provides the internal foundation for Σ_1 , where approximate projectability is treated as a controlled regime condition. In the language of the present paper, approximate projectability is the mathematical expression of physical isolation. A subsystem admits a quantum-effective description when fibre-sensitive ontic variation remains below operational resolution over the relevant timescale.

1.6 Structure of the Paper

Section 2 defines the ontic substrate Σ , including its Kähler structure, Liouville measure, and deterministic Hamiltonian flow.

Section 3 discusses the projection

$$\pi : \Sigma \rightarrow \mathbb{CP}^{N-1}$$

and clarifies the status of quantum states as epistemic images rather than ontic primitives.

Section 4 introduces Σ -level primitives, including Σ -time, Σ -energy, Σ -metric distance, and the possible Σ -level causal rate.

Section 5 defines measurement as de-isolation and relates this to apparatus coupling, outcome sectors, and Born-form operational statistics.

Section 6 states the Bell-facing position of the framework: deterministic, contextual, nonlocal relative to spacetime, and constrained by operational no-signalling.

Section 7 discusses finiteness, bounded resolution, and the cautious status of Planck-scale and cosmological-scale bounds.

Section 8 places Σ_0 within the existing CSD dependency structure.

Section 9 records the principal research targets created by this ontological foundation.

Section 10 summarises the scope and limitations of the paper.

2 The Ontic Substrate Σ

2.1 Ontic State Space

The fundamental state space of CSD is the ontic manifold

$$\Sigma.$$

A point

$$x \in \Sigma$$

represents the actual microstate of the system. This state is not identified with a Hilbert-space vector, a projective ray, or a density operator. Those objects belong to the epistemic or operational description.

In the present paper, Σ is assumed to be a smooth, compact, connected, finite-dimensional Kähler manifold. Thus it is equipped with the structure

$$(\Sigma, \omega_\Sigma, J_\Sigma, g_\Sigma),$$

where ω_Σ is a symplectic form, J_Σ is a compatible complex structure, and g_Σ is the associated Riemannian metric.

The compactness assumption ensures that the total Liouville volume is finite. The connectedness assumption prevents the ontic space from being decomposed into disconnected superselection-like sectors at the foundational level. The finite-dimensional assumption reflects the current scope of the CSD programme and avoids treating infinite-dimensional field structure as primitive.

2.2 Kähler Structure

The symplectic form satisfies

$$d\omega_\Sigma = 0,$$

and is non-degenerate in the sense that

$$\omega_\Sigma(u, v) = 0 \text{ for all } v \in T_x \Sigma \implies u = 0.$$

The complex structure satisfies

$$J_\Sigma^2 = -\text{id},$$

and is compatible with ω_Σ . The metric is defined by

$$g_\Sigma(u, v) = \omega_\Sigma(u, J_\Sigma v).$$

This Kähler package is not derived in this paper. It is adopted as the minimal geometric structure compatible with the downstream use of Hamiltonian flow, invariant volume, and projection to geometric quantum state space.

The motivation is structural. Finite-dimensional geometric quantum mechanics represents pure quantum states by the Kähler manifold $\mathbb{CP}^{N-1}$, equipped with its Fubini-Study metric and symplectic form. CSD treats that projective geometry as epistemic rather than fundamental. It is therefore natural, though not yet derived, to posit a deeper Kähler structure on Σ from which the projective geometry can arise by projection.

2.3 Liouville Measure

Let

$$\dim_{\mathbb{R}} \Sigma = 2d.$$

The symplectic form defines the Liouville volume form

$$\Omega_{\Sigma} = \frac{\omega_{\Sigma}^d}{d!}.$$

The corresponding measure is denoted by

$$\mu_L.$$

Since Σ is compact,

$$\mu_L(\Sigma) < \infty.$$

This finiteness is essential for normalised volume ratios. If $\Omega_0 \subseteq \Sigma$ is a measurable preparation region with

$$0 < \mu_L(\Omega_0) < \infty,$$

then the normalised ontic weight of an outcome region $\Omega_i \subseteq \Omega_0$ is

$$p_i = \frac{\mu_L(\Omega_i)}{\mu_L(\Omega_0)}.$$

This expression is not yet the Born rule. It is the ontic volume-ratio layer. The Born-form expression arises only after the projection to epistemic state space and the additional finite-dimensional measure and operational consistency results developed elsewhere in the CSD programme.

2.4 Deterministic Hamiltonian Dynamics

Ontic evolution is deterministic. Given a smooth Hamiltonian

$$H_{\Sigma} : \Sigma \rightarrow \mathbb{R},$$

the corresponding Hamiltonian vector field $X_{H_{\Sigma}}$ is defined by

$$\iota_{X_{H_{\Sigma}}} \omega_{\Sigma} = dH_{\Sigma}.$$

The ontic trajectory satisfies

$$\frac{dx}{dt_{\Sigma}} = X_{H_{\Sigma}}(x),$$

where t_{Σ} denotes the Σ -level evolution parameter.

The associated flow is written

$$\Phi_t^{\Sigma} : \Sigma \rightarrow \Sigma.$$

Hamiltonian flow preserves the Liouville measure:

$$(\Phi_t^{\Sigma})_* \mu_L = \mu_L.$$

This measure preservation is the structural reason why volume ratios can remain stable under deterministic evolution. It is also the reason why μ_L is the natural measure for the repeated-preparation and volume-typicality layer.

2.5 Ontic and Epistemic Descriptions

The central distinction of the framework is

$$x \in \Sigma \quad \text{ontic,}$$

while

$$[\psi] \in \mathbb{CP}^{N-1} \quad \text{epistemic.}$$

The ontic state x is the actual state of the system. The projective state $[\psi]$ is the accessible quantum-level representation obtained after projection. The epistemic state may therefore identify many distinct ontic states as operationally equivalent.

This distinction prevents the wavefunction from being treated as a primitive physical field. In Σ_0 , the wavefunction is a coordinate representative of a ray in projective space. The ray itself is an epistemic image of deeper ontic structure.

2.6 The Finitist Ontic Posit

The present paper assumes that physically instantiated state structure is finite-dimensional at the foundational level. Infinite-dimensional Hilbert spaces, continuous fields, and continuum spacetime descriptions are therefore treated as effective, limiting, or coarse-grained structures.

This is not a claim that continuum mathematics is invalid. It is a claim about physical ontology. CSD may use continuum methods as mathematical tools while declining to treat physical infinity as primitive.

Accordingly, the current ontic posit is

$$\Sigma \text{ compact and finite-dimensional,}$$

with continuum-like behaviour to be recovered, where possible, from controlled sequences of finite-dimensional models.

2.7 Status of the Ontic Posit

The existence and structure of Σ are not derived in this paper. The paper assumes Σ because the existing finite-dimensional CSD architecture requires an ontic state space with deterministic flow, invariant measure, and a projection to epistemic quantum state space.

The present status is therefore:

$$\Sigma \text{ is a foundational posit.}$$

The corresponding research target is to derive or constrain Σ from deeper principles, such as symmetry, information loss, constrained Hamiltonian structure, finite-resolution state spaces, or a more primitive theory of physical distinguishability.

Thus, Σ_0 does not close the ontic-origin problem. It makes that problem explicit.

3 The Epistemic Projection $\pi : \Sigma \rightarrow \mathbb{CP}^{N-1}$

3.1 Projection to Quantum State Space

CSD assumes a smooth, surjective, many-to-one projection

$$\pi : \Sigma \rightarrow \mathbb{CP}^{N-1}.$$

This map sends an ontic microstate

$$x \in \Sigma$$

to an epistemic projective state

$$\pi(x) = [\psi] \in \mathbb{CP}^{N-1}.$$

The projection is many-to-one. Thus, distinct ontic states may have the same quantum-level representation:

$$x \neq y, \quad \pi(x) = \pi(y).$$

Such states are distinct at the ontic level but indistinguishable at the relevant epistemic quantum level.

The fibre over a ray $[\psi]$ is

$$F_{[\psi]} = \pi^{-1}([\psi]).$$

The fibre contains ontic degrees of freedom not resolved by the projective quantum description.

3.2 Ontic States and Epistemic Rays

The distinction between x and $[\psi]$ is foundational. The ontic state x is the actual physical state. The epistemic ray $[\psi]$ is the projected quantum representation available within a given operational regime.

Thus,

$$x \in \Sigma$$

is not a hidden copy of the wavefunction. It is a deeper state variable.

Likewise,

$$[\psi] \in \mathbb{CP}^{N-1}$$

is not treated as the complete physical state. It is the image of x after projection.

The relation is therefore

$$x \xrightarrow{\pi} [\psi].$$

This structure allows CSD to retain the empirical success of projective quantum state space while denying that projective quantum state space is ontologically fundamental.

3.3 Status of the Projection Map

In this paper, π is primitive structural data. It is assumed, not derived.

This point is essential. The existing finite-dimensional reconstruction requires a projection from ontic states to projective quantum states, but the present programme has not yet derived this map from deeper principles. The correct status is therefore:

$$\pi \text{ is a foundational posit in } \Sigma_0.$$

A stronger future theory would derive π from a physically motivated equivalence relation on Σ . Ideally, one would obtain

$$\mathbb{CP}^{N-1} \cong \Sigma / \sim,$$

where $x \sim y$ means that x and y are indistinguishable under the relevant operational constraints.

At present, this quotient construction remains a research target.

3.4 Compatibility with Fubini-Study Geometry

The epistemic target space $\mathbb{CP}^{N-1}$ carries its standard Fubini-Study structure:

$$(\mathbb{CP}^{N-1}, \omega_{FS}, J_{FS}, g_{FS}, \mu_{FS}).$$

For CSD to reproduce finite-dimensional quantum mechanics, the projection must be compatible with this structure in the quantum-effective sector. In particular, the pushforward of the ontic Liouville measure should take the form

$$\pi_* \mu_L = c \mu_{FS},$$

for some constant $c > 0$, under the relevant symmetry-compatible assumptions.

More generally, when the pushed-forward preparation measure is absolutely continuous with respect to μ_{FS} , one may write

$$\pi_* \mu_L = \rho_{\text{ep}} \mu_{FS},$$

where ρ_{ep} is an epistemic density.

This paper does not rederive the measure bridge. It records why such a bridge is ontologically necessary. Without a controlled relation between μ_L and μ_{FS} , volume structure on Σ would not determine operational quantum weights.

3.5 Projectability of Dynamics

The projection must also interact correctly with dynamics. Let

$$\Phi_t^\Sigma : \Sigma \rightarrow \Sigma$$

be the ontic Hamiltonian flow. A projected flow on $\mathbb{CP}^{N-1}$ is well defined when the evolution of $\pi(x)$ depends only on $\pi(x)$, not on the choice of representative x within the fibre.

In the exact case, this requires

$$\pi(x) = \pi(y) \implies \pi(\Phi_t^\Sigma(x)) = \pi(\Phi_t^\Sigma(y)).$$

Equivalently, the following diagram commutes:

$$\Sigma @>\Phi_t^\Sigma>> \Sigma @V\pi VV @VV\pi V \mathbb{CP}^{N-1} @>\Phi_t^{\text{ep}}>> \mathbb{CP}^{N-1}.$$

Here Φ_t^{ep} is the induced epistemic flow.

This exact condition is too strong for all physical dynamics. The more realistic condition is approximate projectability: fibre-sensitive components of the ontic dynamics remain below operational resolution over the relevant timescale.

3.6 Exact Projectability

A simple sufficient condition for exact projectability is that the ontic Hamiltonian factors through π :

$$H_\Sigma = h \circ \pi,$$

where

$$h : \mathbb{CP}^{N-1} \rightarrow \mathbb{R}.$$

In this case, the Hamiltonian is constant along fibres:

$$x, y \in F_{[\psi]} \implies H_\Sigma(x) = H_\Sigma(y).$$

When h is induced by a self-adjoint operator $\hat{H}$, it takes the standard geometric quantum form

$$h([\psi]) = \frac{\langle \psi | \hat{H} | \psi \rangle}{\langle \psi | \psi \rangle}.$$

The corresponding projected Hamiltonian flow on $\mathbb{CP}^{N-1}$ recovers Schrödinger-form evolution in the finite-dimensional quantum-effective sector.

3.7 Approximate Projectability

Exact projectability is an idealisation. The physically relevant condition is that fibre-sensitive variation is small enough to be operationally invisible over the relevant timescale.

One may write the ontic Hamiltonian schematically as

$$H_\Sigma = h \circ \pi + \delta H_{\text{fib}},$$

where δH_{fib} denotes the fibre-sensitive component.

Approximate projectability requires that

$$\|\delta H_{\text{fib}}\|_{\mathcal{R}, T} \leq \varepsilon,$$

where $\mathcal{R}$ is the operational regime, T is the relevant timescale, and ε is below the resolution threshold for the experiment.

In this interpretation, approximate projectability is the mathematical expression of physical isolation. A subsystem admits a closed quantum-effective description only while fibre-sensitive coupling remains sufficiently suppressed.

3.8 Projection and Information Loss

The projection π is also an information-loss map. It discards ontic distinctions within each fibre:

$$F_{[\psi]} = \pi^{-1}([\psi]).$$

This does not imply subjective ignorance alone. The claim is structural: the operational quantum description does not resolve every ontic degree of freedom.

The epistemic state $[\psi]$ therefore represents an equivalence class of ontic states, not a complete microscopic specification.

This is why CSD can be deterministic at the ontic level while retaining ordinary quantum probabilities at the operational level. The probabilities arise from volume structure over unresolved ontic degrees of freedom after projection and measurement-context selection.

3.9 Open Derivation Problem

The deepest unresolved issue in this section is the origin of π .

The programme requires a future derivation showing why the quotient or coarse-grained image of Σ has the specific structure

$$\mathbb{CP}^{N-1}.$$

A satisfactory derivation should explain:

1. why operational pure states form complex projective space;
2. why the Fubini-Study geometry appears;
3. why the relevant symmetry group is $SU(N)$;
4. why ontic fibres correspond to operationally inaccessible distinctions;
5. why the induced dynamics becomes approximately quantum-effective in isolated regimes.

Until such a derivation is supplied, π remains a controlled foundational posit rather than a completed result.

4 Σ -Level Primitives and Their Spacetime Images

4.1 Why Σ -Level Primitives Are Needed

If Σ is the ontic state space, then several quantities normally treated as spacetime primitives must be reinterpreted. Time, distance, energy, causal rate, and measurement location cannot simply be assumed to originate in spacetime. They must either be primitive at the Σ -level or emerge as coarse-grained images of deeper Σ -level structure.

The working hierarchy of the present paper is therefore

$$\Sigma\text{-level structure} \longrightarrow \text{epistemic quantum structure} \longrightarrow \text{emergent spacetime structure}.$$

This hierarchy is not a derivation of spacetime. It is an ontological ordering. The paper assumes that the Σ -level description is prior, while spacetime descriptions are effective representations valid in appropriate regimes.

4.2 Σ -Time

The Hamiltonian flow on Σ is parameterised by a Σ -level evolution parameter

$$t_\Sigma.$$

An ontic trajectory is written

$$x(t_\Sigma) = \Phi_{t_\Sigma}^\Sigma(x_0).$$

The parameter t_Σ should not automatically be identified with laboratory time t . In the present paper, laboratory time is treated as an emergent or coarse-grained image of the deeper evolution parameter.

Schematically, one may write

$$t = C_t(t_\Sigma, x, \mathcal{R}),$$

where C_t denotes an unspecified reconstruction or coarse-graining map, and $\mathcal{R}$ denotes the operational regime.

This notation is deliberately schematic. Its purpose is to prevent an illicit identification between Σ -time and spacetime time before the relationship has been derived.

4.3 Σ -Energy

The ontic Hamiltonian

$$H_\Sigma : \Sigma \rightarrow \mathbb{R}$$

generates deterministic motion on Σ . Operational energy is then interpreted as the projected or coarse-grained image of this deeper generator.

In an exactly projectable regime, one has

$$H_\Sigma = h \circ \pi,$$

where

$$h : \mathbb{CP}^{N-1} \rightarrow \mathbb{R}.$$

If h is induced by a self-adjoint operator $\hat{H}$, then

$$h([\psi]) = \frac{\langle \psi | \hat{H} | \psi \rangle}{\langle \psi | \psi \rangle}.$$

In this case, ordinary quantum energy is recovered as the projective expression of Σ -level Hamiltonian structure.

Outside the exact regime, the Hamiltonian may contain fibre-sensitive terms:

$$H_\Sigma = h \circ \pi + \delta H_{\text{fib}}.$$

The term δH_{fib} represents ontic variation not resolved by the epistemic quantum state. Such terms are compatible with a quantum-effective description only when they remain sufficiently small over the relevant operational timescale.

4.4 Σ -Metric Distance

The metric g_Σ defines distance on the ontic manifold. Let

$$D_\Sigma(x, y)$$

denote the metric distance between two ontic states $x, y \in \Sigma$.

This distance need not correspond directly to spacetime distance. If spacetime is emergent, then spacetime distance must be reconstructed from Σ -level data.

Let

$$C_{\text{st}} : \Sigma \rightarrow M_{\text{st}}$$

denote a future coarse-graining map from ontic states to an effective spacetime description. Then the spacetime distance between two images is schematically

$$D_{\text{st}}(C_{\text{st}}(x), C_{\text{st}}(y)).$$

There is no general assumption that

$$D_{\text{st}}(C_{\text{st}}(x), C_{\text{st}}(y))$$

is a monotonic function of

$$D_\Sigma(x, y).$$

Indeed, CSD may require coarse-graining maps that stretch Σ -near relations into spacetime-distant relations. This possibility is central to the later interpretation of Bell-type correlations.

4.5 Causal Rate in Σ

The paper introduces a Σ -level maximum causal rate,

$$c_\Sigma.$$

This quantity is not automatically identified with the observed speed of light c . The cautious relation is

$$c = C_c(c_\Sigma, \mathcal{R}),$$

where C_c denotes the regime-dependent map from Σ -level causal propagation to effective spacetime causal structure.

A stronger theory would show either

$$c_\Sigma = c,$$

or derive c as an emergent limit of c_Σ under coarse-graining.

The present paper does neither. It records c_Σ as a required primitive or intermediate quantity for future spacetime-facing work.

4.6 Emergent Spacetime

The emergence of spacetime requires a map, or family of maps, of the form

$$C_{\text{st}} : \Sigma \rightarrow M_{\text{st}},$$

where M_{st} denotes an effective spacetime manifold or spacetime-like structure.

This map is not the same as the epistemic projection

$$\pi : \Sigma \rightarrow \mathbb{CP}^{N-1}.$$

The projection π maps ontic states to quantum state descriptions. The map C_{st} would map ontic structure to spacetime structure.

The distinction is:

$$\pi : \Sigma \rightarrow \mathbb{CP}^{N-1} \quad \text{quantum epistemic projection,}$$

whereas

$$C_{\text{st}} : \Sigma \rightarrow M_{\text{st}} \quad \text{spacetime coarse-graining.}$$

Conflating these maps would be a category error. The first concerns quantum state representation. The second concerns the emergence of spacetime localisation, metric structure, and causal order.

4.7 Stretched Coarse-Graining

One possible route to spacetime nonlocality is stretched coarse-graining. The basic idea is that states close in Σ may have images that are distant in spacetime:

$$D_{\Sigma}(x, y) \text{ small,}$$

while

$$D_{\text{st}}(C_{\text{st}}(x), C_{\text{st}}(y)) \text{ large.}$$

If this occurs, then correlations that appear nonlocal in spacetime may reflect local or adjacent structure in Σ .

This statement is speculative at the present stage. It is not a theorem. It is a research direction that may become relevant when CSD is extended to effective spacetime and locality.

4.8 Operational Regimes

All spacetime-facing identifications in this section are regime-dependent. A laboratory system does not access Σ directly. It accesses stable coarse-grained structures that behave as if they occupy spacetime, evolve in time, and interact through effective causal constraints.

Thus, an operational regime $\mathcal{R}$ must specify at least:

1. the relevant projection to quantum state space;
2. the relevant coarse-graining to spacetime variables;
3. the experimental resolution scale;
4. the timescale over which approximate projectability holds;
5. the class of observables accessible to the apparatus.

Without specifying $\mathcal{R}$, statements about time, distance, energy, and locality are incomplete.

4.9 Summary

The quantities t_{Σ} , H_{Σ} , D_{Σ} , and c_{Σ} are introduced as Σ -level primitives or intermediate structures. Their spacetime counterparts are not assumed to be identical to them.

The current status is therefore:

$$t_{\Sigma}, H_{\Sigma}, D_{\Sigma}, c_{\Sigma} \text{ belong to the ontic description,}$$

while

$$t, H, D_{\text{st}}, c \text{ belong to the emergent spacetime description.}$$

The relationship between these two levels remains one of the principal open problems for the Σ -series.

5 Measurement as Σ -Level De-Isolation

5.1 Measurement as a Physical Transition

In CSD, measurement is not treated as a primitive stochastic event. It is a physical transition in which a previously isolated or approximately isolated subsystem becomes dynamically coupled to apparatus, environment, or record-forming degrees of freedom.

Before measurement, a subsystem may admit an approximately closed quantum-effective description:

$$x_{\text{sys}} \in \Sigma_{\text{sys}}, \quad \pi_{\text{sys}} : \Sigma_{\text{sys}} \rightarrow \mathbb{CP}^{N-1}.$$

During measurement, the relevant ontic system is enlarged:

$$\Sigma_{\text{sys}} \longrightarrow \Sigma_{\text{SA}},$$

where Σ_{SA} denotes the joint system-apparatus ontic manifold.

The measurement event is therefore a de-isolation event. The subsystem no longer evolves as an autonomous approximately projectable sector. Its later behaviour is determined by the joint dynamics on Σ_{SA} .

5.2 Physical Isolation and Approximate Projectability

A subsystem is physically isolated when its projected epistemic dynamics are approximately closed over the relevant operational timescale. In the exact idealisation, if two ontic states lie in the same fibre,

$$\pi(x) = \pi(y),$$

then their projected futures agree:

$$\pi(\Phi_t^\Sigma(x)) = \pi(\Phi_t^\Sigma(y)).$$

In practice, exact projectability is too strong. The appropriate condition is approximate projectability. If

$$H_\Sigma = h \circ \pi + \delta H_{\text{fib}},$$

then approximate projectability requires the fibre-sensitive component δH_{fib} to remain below operational resolution:

$$\|\delta H_{\text{fib}}\|_{\mathcal{R}, T} \leq \varepsilon.$$

Here $\mathcal{R}$ is the operational regime, T is the relevant timescale, and ε is the tolerated error scale.

Thus, in the language of this paper,

$$\text{physical isolation} \leftrightarrow \text{approximate projectability}.$$

This identification is interpretive in Σ_0 . Its controlled mathematical treatment belongs to Σ_1 .

5.3 De-Isolation and Apparatus Coupling

Measurement begins when the isolated subsystem approximation fails because the system becomes coupled to additional degrees of freedom. The joint Hamiltonian may be written schematically as

$$H_{\text{SA}} = H_{\text{sys}} + H_{\text{app}} + H_{\text{int}},$$

where H_{int} is the system-apparatus interaction term.

When H_{int} is non-negligible over the measurement timescale, the autonomous subsystem description is no longer sufficient. The correct ontic trajectory is instead

$$x_{\text{SA}}(t_\Sigma) \in \Sigma_{\text{SA}}.$$

The projected quantum description also changes. Instead of a closed subsystem ray, one obtains a joint system-apparatus description:

$$\pi_{\text{SA}} : \Sigma_{\text{SA}} \rightarrow \mathbb{CP}^{N_{\text{SA}}-1}.$$

The apparent measurement outcome arises from the joint dynamics, not from a stochastic collapse imposed on the subsystem alone.

5.4 Outcome Regions

A measurement context M determines outcome regions. At the ontic level, these are measurable subsets

$$\Omega_i^\Sigma(M) \subseteq \Sigma_{\text{SA}}.$$

At the epistemic level, their projected counterparts are measurable subsets

$$\Omega_i^{\text{ep}}(M) \subseteq \mathbb{CP}^{N_{\text{SA}}-1}.$$

The outcome regions must satisfy:

$$\Omega_i^\Sigma(M) \cap \Omega_j^\Sigma(M) = \emptyset \quad \text{for } i \neq j,$$

and, up to null boundaries,

$$\bigcup_i \Omega_i^\Sigma(M) = \Omega_M^\Sigma,$$

where Ω_M^Σ is the measurement-relevant region of the joint ontic manifold.

The realised outcome is determined by the actual ontic state after the measurement interaction:

$$x_{\text{SA}}(T) \in \Omega_i^\Sigma(M) \implies \text{outcome } i.$$

Thus, outcome definiteness is not added by collapse. It follows from the existence of an actual ontic trajectory and a context-defined partition of the relevant joint state space.

5.5 Macroscopic Records

A laboratory measurement record is not a primitive ontological object. It is a stable coarse-grained image of many lower-level ontic interactions.

A detector click, pointer position, screen mark, or memory record corresponds to a robust macroscopic sector. Schematically:

$$\Sigma\text{-level joint dynamics} \longrightarrow \text{stable pointer sector} \longrightarrow \text{spacetime record}.$$

The apparatus is therefore treated as a physical system whose stable records arise from deterministic dynamics on a larger ontic manifold.

This is important because CSD does not require observers to play a primitive role. Observers are physical systems embedded in the same ontic structure as the measured system and apparatus.

5.6 No Primitive Collapse

CSD does not add a collapse postulate. The transition from a pre-measurement description to an outcome-conditioned post-measurement description is interpreted as conditionalisation on the realised sector.

At the ontic level, there is a single realised trajectory:

$$x_{\text{SA}}(t_{\Sigma}).$$

At the epistemic or operational level, after outcome i is recorded, the relevant description is restricted to the sector compatible with that outcome:

$$\Omega_i^{\Sigma}(M).$$

In ideal projective cases, this sector restriction is represented operationally by Lüders-type update. In generalised measurements, it is represented by the corresponding instrument-level update.

The update is therefore not a stochastic physical jump at the ontic level. It is the operational representation of deterministic evolution plus outcome-conditioned restriction.

5.7 Born-Form Statistics

The Born rule is not introduced at the measurement stage as a primitive probability law. The statistical chain is layered.

At the ontic level, one has volume ratios:

$$p_i = \frac{\mu_L(\Omega_i^{\Sigma})}{\mu_L(\Omega_0^{\Sigma})},$$

where Ω_0^{Σ} is the relevant preparation region and Ω_i^{Σ} is an outcome sector.

After projection to the epistemic layer, and under the finite-dimensional measure and operational consistency conditions established elsewhere in the CSD programme, these weights are represented in Born form:

$$p_i = \langle \psi | \hat{P}_i | \psi \rangle$$

for projective measurements, and

$$p_i = \text{Tr}(\rho \hat{E}_i)$$

for generalised measurements.

Thus, Born-form probabilities are operational representations of the projected volume structure. They are not primitive stochastic propensities in Σ_0 .

5.8 Measurement Context Dependence

Outcome regions are context-dependent. Changing the measurement arrangement changes the system-apparatus coupling, the pointer sectors, or the relevant partition of the joint state space.

Thus, for two contexts M and M' , there is no general reason to expect

$$\Omega_i^{\Sigma}(M) = \Omega_i^{\Sigma}(M').$$

This context dependence is not an optional interpretive addition. It is required if deterministic CSD is to remain compatible with Bell and Kochen-Specker constraints.

The theory assigns a definite outcome for the context actually implemented. It does not assign a single global set of outcomes for all incompatible measurement contexts simultaneously.

5.9 Observers as Ordinary Sigma-Structures

CSD treats observers as ordinary physical systems. They are not granted a privileged ontological role, and they do not enter the formalism as primitive elements.

An observer is a region of Σ , or more precisely a coarse-grained structure built from many Σ -level degrees of freedom, evolving under the same deterministic Hamiltonian flow as any other physical system. Observers, apparatuses, environments, and measured systems all live on the same ontic manifold and are subject to the same dynamics

$$\Phi_t^\Sigma : \Sigma \rightarrow \Sigma.$$

There is no separate observer category, no conscious-collapse mechanism, and no requirement that measurement records be accessed by a perceiving subject in order to count as physically realised.

The consequences are threefold.

First, the von Neumann measurement chain has no canonical termination point in CSD. A measurement record exists as soon as the relevant stable pointer sector has been realised by the joint Σ_{SA} dynamics. Whether that record is subsequently read by a human, a machine, or no one at all has no ontological significance.

Second, the standard objections to observer-dependent quantum interpretations do not apply. CSD makes no claim that physical quantities require observation to be definite, nor that the laws of physics depend on the presence of agents.

Third, observer-system distinctions are operational and contextual rather than fundamental. In one experiment, a photodetector is the apparatus and a human experimenter is the observer. In a different experiment, the experimenter's retina and visual cortex are the apparatus and the experimenter is the observer of their own internal record. In either case, the underlying Σ -level description is the same: a joint ontic trajectory passing through context-defined outcome sectors. The split between "system," "apparatus," and "observer" is a description choice, not a physical partition.

This flat ontology dissolves the measurement-problem asymmetry that requires standard quantum mechanics to treat observers or classical apparatuses as ontologically distinct. In CSD, the asymmetry is replaced by the structural distinction between approximately projectable subsystems and the joint Σ_{SA} dynamics into which they are de-isolated. That distinction is dynamical, not categorical.

5.10 Summary

Measurement in CSD is a deterministic physical process on a joint ontic manifold. It consists of:

1. approximate subsystem isolation before measurement;
2. de-isolation through system-apparatus coupling;
3. deterministic evolution on Σ_{SA} ;
4. context-defined outcome sectors;
5. realisation of one sector by the actual ontic trajectory;
6. operational update by conditioning on the realised sector.

The central formula is therefore not a collapse rule, but the transition

$$\Sigma_{\text{sys}} \longrightarrow \Sigma_{SA} \longrightarrow \{\Omega_i^\Sigma(M)\}.$$

This is the Σ -level structure underlying measurement in the present ontology.

6 Bell Compatibility, Contextuality, and Nonlocality

6.1 The Bell Constraint

Any deterministic account of quantum phenomena must state its relationship to Bell-type constraints explicitly. If a theory reproduces Bell-violating quantum correlations, it cannot also retain Bell-local outcome functions defined on a single hidden-variable space.

The excluded structure is the factorised form

$$P(s, t|a, b, \lambda) = P_A(s|a, \lambda)P_B(t|b, \lambda),$$

together with a preparation distribution independent of the later measurement settings.

In deterministic form, this would amount to outcome functions

$$s = A(a, \lambda), \quad t = B(b, \lambda),$$

defined for all relevant settings on one common probability space.

CSD does not assume this structure. It is deterministic, but it is not Bell-local in the standard spacetime sense.

6.2 Context-Indexed Outcome Structure

In CSD, a measurement context determines the relevant system-apparatus interaction and the corresponding outcome sectors. For a Bell experiment with detector settings a and b , the outcome sectors are context-indexed:

$$\Omega_{st}^\Sigma(a, b) \subseteq \Sigma_{SA},$$

where

$$s, t \in \{+1, -1\}.$$

The realised outcome is definite for the actually implemented context:

$$x_{SA}(T) \in \Omega_{st}^\Sigma(a, b) \implies (s, t).$$

However, CSD does not require a single global map assigning definite outcomes for every incompatible setting simultaneously. That is, it does not require a single object of the form

$$(A(a, \lambda), A(a', \lambda), B(b, \lambda), B(b', \lambda))$$

for all incompatible choices a, a', b, b' .

This distinction is essential. CSD asserts deterministic outcomes for performed measurements. It does not assert global preassignment of outcomes for all unperformed incompatible measurements.

6.3 Contextual Determinism

The appropriate description is contextual determinism.

For each realised context M , there is a deterministic outcome map

$$R_M : \Sigma_{SA} \rightarrow \mathcal{O}_M,$$

where $\mathcal{O}_M$ is the finite outcome set for that context.

For a Bell context $M = (a, b)$, this becomes

$$R_{a,b} : \Sigma_{SA} \rightarrow \{+1, -1\} \times \{+1, -1\}.$$

Thus,

$$R_{a,b}(x_{SA}) = (s, t)$$

is well defined for the context actually implemented.

The framework does not require that

$$R_{a,b}, \quad R_{a,b'}, \quad R_{a',b}, \quad R_{a',b'}$$

be restrictions of one common global outcome assignment. Indeed, if Bell-violating correlations are reproduced, such a global assignment cannot exist.

6.4 Nonlocality Relative to Emergent Spacetime

CSD should be described as nonlocal relative to emergent spacetime. This is the conservative and correct statement.

In a Bell experiment, the two detector outcomes may be spacetime-separated. If the joint outcome sectors depend on the full measurement context (a, b) , then the framework is not local in the Bell sense.

The relevant point is not that a signal travels from one detector to the other through spacetime. Rather, the joint outcome structure is defined at the level of the full system-apparatus configuration. The underlying ontic state belongs to a joint space, and the outcome sectors are determined by the realised joint context.

Thus, the spacetime description may display nonlocal correlation, while the Σ -level description may involve a single structured joint event.

6.5 Σ -Locality as an Open Possibility

Although CSD is nonlocal relative to emergent spacetime, it leaves open the possibility that the relevant structure is local or adjacent in Σ .

Let

$$D_{\Sigma}(x, y)$$

denote ontic distance, and let

$$C_{\text{st}} : \Sigma \rightarrow M_{\text{st}}$$

be a future spacetime coarse-graining map. It may be possible that

$$D_{\Sigma}(x, y)$$

is small while

$$D_{\text{st}}(C_{\text{st}}(x), C_{\text{st}}(y))$$

is large.

If so, Bell-type correlations that appear nonlocal in spacetime may arise from structurally close or jointly constrained regions of Σ .

This is not claimed as a theorem in the present paper. It is a research target for later spacetime-facing work. The safe conclusion is narrower:

CSD is nonlocal in spacetime, but may be local in a deeper Σ -level sense.

6.6 No Operational Signalling

Contextual nonlocality must not permit controllable superluminal signalling. The required operational condition is marginal stability.

For detector A,

$$P_A(s|a, b) = P_A(s|a, b')$$

for all relevant b, b' . For detector B,

$$P_B(t|a, b) = P_B(t|a', b)$$

for all relevant a, a' .

Equivalently,

$$P_A(s|a, b) = P_A(s|a),$$

and

$$P_B(t|a, b) = P_B(t|b).$$

In the singlet case, the operational marginals take the form

$$P_A(s|a, b) = \frac{1}{2}, \quad P_B(t|a, b) = \frac{1}{2}.$$

Thus, the dependence of joint outcome structure on the context (a, b) does not by itself imply signalling. Signalling would require controllable variation of a local marginal by changing the remote setting. CSD must exclude that at the operational level.

6.7 Joint Correlations

For the two-qubit singlet sector, the standard quantum correlation is

$$E(a, b) = -a \cdot b.$$

This can be written as

$$E(a, b) = \sum_{s, t = \pm 1} st P_{st}(a, b).$$

The corresponding joint probabilities are

$$P_{st}(a, b) = \frac{1 - st a \cdot b}{4}.$$

These probabilities violate Bell inequalities for suitable choices of detector settings, while preserving the no-signalling marginals.

Within CSD, the interpretation is that these are operational probabilities arising from the finite-dimensional Born-weight layer and the relevant system-apparatus pointer-sector construction. They are not produced by Bell-local preassigned values.

6.8 Comparison with Bell-Local Hidden Variables

A Bell-local hidden-variable model attempts to explain correlations by introducing a hidden variable λ such that

$$P(s, t|a, b) = \int P_A(s|a, \lambda) P_B(t|b, \lambda) d\rho(\lambda).$$

CSD does not take this form. Its relevant variable is an ontic state in a joint ontic manifold,

$$x_{SA} \in \Sigma_{SA},$$

and the outcome sectors are context-indexed:

$$\Omega_{st}^\Sigma(a, b).$$

The difference is structural. CSD is not trying to recover quantum correlations while preserving Bell-local factorisation. It rejects that factorisation as the wrong mathematical representation of a contextual deterministic theory.

6.9 Relation to Fine's Theorem

Fine's theorem shows that, in the Bell-CHSH setting, satisfaction of the Bell inequalities is equivalent to the existence of a joint probability distribution over all incompatible measurement outcomes.

Since quantum mechanics violates Bell inequalities, a theory reproducing quantum predictions must not imply such a global joint distribution.

CSD avoids this implication by denying the existence of one global assignment across incompatible contexts. It permits deterministic outcome maps for realised contexts:

$$R_{a,b},$$

without requiring a single context-independent map from which all $R_{a,b}$ arise.

Thus, the framework is compatible with Fine-type constraints because it does not assert the structure that Fine's theorem rules out.

6.10 Interpretive Position

The Bell-facing position of Σ_0 can be summarised as follows:

$$\text{deterministic} + \text{contextual} + \text{spacetime-nonlocal} + \text{operationally no-signalling}.$$

This is the correct conservative position for CSD at the present stage.

It would be too strong to claim that CSD has already derived a complete Σ -local mechanism for Bell correlations. It would also be misleading to describe CSD as Bell-local. The appropriate statement is that CSD rejects Bell-local factorisation while preserving deterministic realised outcomes and no operational signalling.

6.11 Summary

CSD is compatible with Bell constraints only because it is not a Bell-local hidden-variable theory. It does not assign global outcomes for all incompatible measurements on a single shared probability space. Instead, each realised measurement context determines its own outcome sectors on the joint system-apparatus manifold.

The resulting picture is:

$$\text{single ontic trajectory} \longrightarrow \text{context-indexed outcome sector} \longrightarrow \text{definite observed result}.$$

Bell-type correlations are therefore not explained by local pre-existing values. They arise from contextual joint structure, finite-dimensional Born-form weights, and deterministic sector realisation.

7 Finiteness, Resolution, and Physical Infinity

7.1 Finiteness as an Ontic Postulate

CSD does not treat physical infinity as a primitive ontological commitment. The present framework assumes that the fundamental state space is compact and finite-dimensional:

$$\dim_{\mathbb{R}} \Sigma < \infty, \quad \mu_L(\Sigma) < \infty.$$

This does not mean that infinite-dimensional mathematics is rejected. It means that infinite-dimensional structures are not assumed to be physically fundamental. They may appear as effective descriptions, limiting constructions, or continuum approximations.

The current ontic stance is therefore:

finite ontic structure first, continuum behaviour later.

This position is consistent with the finite-dimensional scope of the existing CSD reconstruction. It also aligns with the programme's use of controlled truncations, finite N models, and continuum-like comparison only as an asymptotic or coarse-grained layer.

7.2 Compactness and Finite Measure

The compactness of Σ implies finite Liouville measure:

$$\mu_L(\Sigma) < \infty.$$

This matters because volume-based probability requires normalisable preparation regions. If

$$\Omega_0 \subseteq \Sigma$$

is a measurable preparation region, then the expression

$$\frac{\mu_L(\Omega_i)}{\mu_L(\Omega_0)}$$

is meaningful only when

$$0 < \mu_L(\Omega_0) < \infty.$$

Thus, finite measure is not a cosmetic assumption. It is part of the statistical architecture of the theory.

However, compactness of Σ does not by itself imply that emergent spacetime has finite spatial volume, finite temporal extent, or finite information capacity. Those would require additional arguments involving the coarse-graining map from Σ to spacetime.

7.3 Physical Resolution

If spacetime is emergent, then spacetime resolution should not be assumed to be arbitrarily fine. The physically meaningful question is not whether a mathematical continuum can be written down, but whether arbitrarily small spacetime distinctions correspond to distinct physical states.

The working hypothesis is that spacetime resolution is bounded. Schematically,

$$\Delta x_{\min} \leq \Delta x_{\text{st}} \leq \Delta x_{\max},$$

where Δx_{st} denotes an operational spacetime resolution scale.

The lower and upper bounds are not derived in this paper. They are introduced only as possible physical constraints on the emergent spacetime description.

7.4 Planck-Scale Lower Bound

A natural candidate for a lower spacetime resolution scale is the Planck length:

$$\ell_P = \sqrt{\frac{\hbar G}{c^3}}.$$

The cautious CSD statement is:

$$\Delta x_{\text{st}} \gtrsim \ell_P$$

may hold in regimes where spacetime geometry, quantum structure, and gravitational coupling are all relevant.

This is not a derived result of Σ_0 . The paper does not prove that ℓ_P follows from Σ -geometry. It only records the Planck scale as the standard candidate lower bound for physically meaningful spacetime localisation.

A future derivation would need to explain how $\hbar$, G , and c arise from or relate to Σ -level structure.

7.5 Cosmological-Scale Upper Bound

A natural candidate for an upper spacetime scale is set by the cosmological constant. The associated length scale is

$$L_\Lambda \sim \Lambda^{-1/2}.$$

The cautious CSD statement is:

$$\Delta x_{\text{st}} \lesssim L_\Lambda$$

may mark a regime beyond which global spacetime resolution, causal structure, or horizon-scale distinguishability must be treated differently.

Again, this is not derived in Σ_0 . The paper does not prove that L_Λ follows from compactness of Σ , nor that it is the exact upper resolution scale of emergent spacetime.

It is recorded as a possible upper scale for later spacetime-facing work.

7.6 Bounded Spacetime Window

Combining the preceding candidate scales gives the schematic bounded-resolution window

$$\ell_P \lesssim \Delta x_{\text{st}} \lesssim L_\Lambda.$$

This expression should be read as a research hypothesis, not as a theorem.

The present paper does not claim that CSD derives this window. It states only that, if spacetime is emergent from finite ontic structure, then physical resolution should plausibly be bounded below and above.

The lower bound would express the failure of arbitrarily fine spacetime localisation. The upper bound would express the failure of treating arbitrarily large spacetime regions as sharply resolved independent physical structures.

7.7 Finite Ontology Versus Finite Spacetime

It is important to distinguish finite ontic structure from finite spacetime.

The statement

$$\mu_L(\Sigma) < \infty$$

is a statement about the ontic manifold.

It does not automatically imply:

$$\text{Vol}(M_{\text{st}}) < \infty,$$

nor does it imply that spacetime is discrete, compact, closed, or finite in ordinary cosmological terms.

To derive any of those claims, CSD would need a defined map

$$C_{\text{st}} : \Sigma \rightarrow M_{\text{st}},$$

together with a theorem connecting compactness or finite measure in Σ to metric, causal, or volumetric properties of M_{st} .

No such theorem is supplied here.

7.8 Continuum Descriptions as Effective Tools

Continuum models remain useful. A theory may be finite at the ontic level while admitting continuum approximations at effective scales.

The CSD position is therefore not:

continuum mathematics is invalid.

It is:

continuum ontology is not assumed as primitive.

This distinction matters. The existing CSD programme can use Hilbert spaces, projective manifolds, smooth geometry, fields, and limiting procedures as mathematical tools, while still treating finite-dimensional compact ontic structure as foundational.

7.9 Summary

The finitist position of Σ_0 has three parts.

First, Σ is assumed compact and finite-dimensional:

$$\dim_{\mathbb{R}} \Sigma < \infty, \quad \mu_L(\Sigma) < \infty.$$

Second, infinite-dimensional or continuum structures are treated as effective, limiting, or coarse-grained descriptions.

Third, physically meaningful spacetime resolution is expected to be bounded, with

$$\ell_P \lesssim \Delta x_{\text{st}} \lesssim L_{\Lambda}$$

recorded as a candidate research hypothesis rather than a derived CSD result.

Thus, Σ_0 rejects physical infinity as a primitive posit while leaving the technical derivation of spacetime bounds to later work.

8 Relation to the Existing CSD Corpus

8.1 Purpose of This Placement

Σ_0 is not a replacement for any existing CSD paper. It is an internal foundation paper that clarifies what the technical corpus already assumes about the ontic layer.

The existing papers and notes develop specific components of the programme: volume typicality, the measure bridge, Born-form operational weights, finite-dimensional reconstruction, measurement update, mixed states, Bell-sector structure, many-mode models, continuum-like limits, and Lean formalisation.

Σ_0 has a different role. It states the physical interpretation of the objects used across that corpus:

$$\Sigma, \quad x \in \Sigma, \quad \mu_L, \quad \pi : \Sigma \rightarrow \mathbb{CP}^{N-1}, \quad \Omega_i^{\Sigma}, \quad t_{\Sigma}.$$

Its purpose is to prevent hidden ontological assumptions from being mistaken for derived results.

8.2 Relation to Paper A

Paper A establishes the deterministic volume-typicality layer. It shows how repeated preparation on a finite-measure state space can yield long-run outcome frequencies equal to normalised volume weights.

In schematic form, Paper A supplies:

$$\text{deterministic measure-preserving dynamics} \longrightarrow \text{frequency convergence to volume ratios.}$$

Σ_0 supplies the ontological reading of the state space used in that theorem. It identifies the relevant physical state space with Σ , the actual state with $x \in \Sigma$, and the invariant measure with μ_L .

Thus, Paper A supplies the statistical mechanism. Σ_0 supplies the physical interpretation of the mechanism.

8.3 Relation to Paper B

Paper B addresses the transition from volume-based weights to Born-form operational weights. It uses symmetry and operational consistency to constrain the probability assignment on finite-dimensional projective quantum state space.

In schematic form, Paper B contributes:

$$\text{projective state space} + \text{symmetry and operational consistency} \longrightarrow \text{Born-form weights.}$$

Σ_0 does not repeat this argument. It explains why such an argument is needed. If the ontic measure μ_L is to explain quantum statistics, it must connect to the projective quantum measure structure on

$$\mathbb{CP}^{N-1}.$$

Thus, Paper B belongs to the epistemic and operational weight layer, while Σ_0 belongs to the ontic foundation layer.

8.4 Relation to Paper C

Paper C gives the finite-dimensional reconstruction architecture. It introduces the main working structure:

$$\Sigma \xrightarrow{\pi} \mathbb{CP}^{N-1},$$

together with quantum-effective Hamiltonian dynamics, measurement regions, composite structure, and the operational dictionary.

Σ_0 is logically prior to Paper C in one respect: it states what the objects in that architecture are meant to represent.

Paper C uses Σ as a mathematical state space. Σ_0 says that Σ is intended as the ontic state space.

Paper C uses π as the projection to quantum state space. Σ_0 says that π is currently primitive structural data and that its deeper derivation remains open.

Paper C uses outcome regions. Σ_0 interprets them as context-defined sectors selected by deterministic system-apparatus dynamics.

8.5 Relation to Paper D

Paper D presents CSD as an ontological completion of geometric quantum mechanics. It is the public-facing conceptual synthesis of the finite-dimensional programme.

Σ_0 is more internal and more explicit. It can state openly that several structures used by the programme are not yet derived from first principles:

$$\Sigma, \quad \pi, \quad C_{\text{st}}, \quad c_{\Sigma}, \quad \text{emergent spacetime.}$$

This makes Σ_0 a useful support document for Paper D. It protects the public interpretation from overclaiming by recording, internally, which elements remain assumptions or research targets.

The correct relationship is therefore:

$$\Sigma_0 \text{ clarifies the internal ontology,}$$

while

$$\text{Paper D} \text{ presents the public ontological synthesis.}$$

8.6 Relation to TN0

TN0 fixes canonical notation and conventions across the programme. Σ_0 should follow TN0 throughout.

In particular, the ontic microstate is written

$$x \in \Sigma,$$

not $\omega \in \Sigma$.

The ontic measure is written

$$\mu_L.$$

The epistemic measure is written

$$\mu_{FS}.$$

Ontic outcome regions should be denoted

$$\Omega_i^\Sigma,$$

while epistemic outcome regions should be denoted

$$\Omega_i^{\text{ep}}.$$

This is important because Σ_0 is intended to sit at the base of the later Σ -series. It should not preserve older notation that TN0 has already superseded.

8.7 Relation to TN1

TN1 supplies the measure-bridge support note. It establishes, under symmetry-compatible assumptions, that the pushforward of Liouville measure is proportional to the Fubini-Study measure:

$$\pi_*\mu_L = c\mu_{FS}.$$

Σ_0 does not prove this theorem. It states why the theorem matters ontologically.

Without a bridge of this kind, the invariant volume structure on Σ would not automatically explain the canonical projective measure structure used in finite-dimensional quantum mechanics.

Thus, TN1 closes a technical measure gap. Σ_0 identifies the ontological role of that closure.

8.8 Relation to TN2 and TN3

TN2 and TN3 address the Bell-singlet sector.

TN2 gives the non-constructive finite-dimensional singlet specialisation of the Born-weight layer. TN3 gives the constructive quantum-pointer Hamiltonian realisation of the Bell-singlet correlation.

Σ_0 supplies the interpretive frame for these results. It states that CSD should be understood as:

deterministic, contextual, spacetime-nonlocal, operationally no-signalling.

Thus, TN2 and TN3 provide Bell-sector technical content, while Σ_0 states the corresponding ontological position.

8.9 Relation to TN4

TN4 develops the finite-dimensional operational dictionary for mixed states, POVMs, and subsystem reduction.

In Σ_0 , density operators and POVMs are not ontic primitives. They are operational representation objects.

The ontic structure remains:

$$x \in \Sigma, \quad \mu_L, \quad \Phi_t^\Sigma.$$

The operational structure is:

$$\rho, \quad \{\hat{E}_i\}, \quad p_i = \text{Tr}(\rho\hat{E}_i).$$

TN4 explains how the latter can arise from the former in the finite-dimensional setting. Σ_0 clarifies that the latter should not be mistaken for fundamental ontology.

8.10 Relation to TN6

TN6 develops post-measurement update and sequential experiments.

Σ_0 interprets this layer as follows. At the ontic level, there is no primitive collapse. There is one deterministic joint trajectory:

$$x_{\text{SA}}(t_\Sigma) \in \Sigma_{\text{SA}}.$$

Operational update occurs because, after an outcome is recorded, the description is restricted to the realised outcome sector:

$$\Omega_i^\Sigma(M).$$

TN6 supplies the technical sequential-update machinery. Σ_0 supplies the ontological reading: update is conditioned description after deterministic sector realisation, not stochastic physical collapse.

8.11 Relation to TN7

TN7 extends the finite-dimensional architecture to coupled finite-mode systems and joint ontic structure.

Σ_0 supports TN7 by stating that composite systems should not be assumed to arise from naive Cartesian factorisation at the ontic level. A joint system may require a non-factorising ontic manifold:

$$\Sigma_{AB} \not\cong \Sigma_A \times \Sigma_B.$$

This point is important for entanglement, marginal stability, and Bell-sector compatibility.

TN7 develops the coupled finite-mode machinery. Σ_0 records the ontological reason why joint structure is fundamental rather than secondary.

8.12 Relation to c1 and c2

The companion paper c1 addresses Fine-Bell consistency. It clarifies why deterministic realised outcomes do not imply global joint outcome assignments for incompatible measurement settings.

The companion paper c2 addresses PBR compatibility. It clarifies why preparation independence, understood as product factorisation over independently prepared subsystems, does not straightforwardly apply to non-factorising CSD composite ontology.

Σ_0 provides the deeper ontological background for both papers:

outcomes are context-indexed,

and

composite ontic structure need not factorise.

Thus, c1 and c2 are no-go-facing technical clarifications, while Σ_0 records the ontological commitments that make those clarifications natural.

8.13 Relation to c3

c3 develops controlled continuum-like limits from nested finite truncations.

Σ_0 supplies the ontological reason for this strategy. Since the fundamental state space is assumed finite-dimensional and compact, continuum structures should not be assumed as primitive. They should be recovered, if possible, through controlled limiting or coarse-grained procedures.

Thus, c3 is the technical bridge toward continuum-like behaviour, while Σ_0 states the finitist ontic motivation for needing such a bridge.

8.14 Relation to LF1, LF2, and LF3

The Lean formalisation branch verifies selected theorem layers of the programme.

LF1 formalises repeated-trial frequency convergence.

LF2 formalises the measure-bridge and Born-weight wrapper layer.

LF3 formalises the Bell-singlet pointer-sector layer and its empirical-chain capstones.

Σ_0 is not a Lean formalisation paper. It does not add a machine-checked theorem. Its role is to state what the formalised objects mean physically.

In particular:

LF1 verifies a statistical theorem,

LF2 verifies a measure and weight wrapper,

LF3 verifies a Bell-sector kernel,

while

Σ_0 states the ontological interpretation of the formal chain.

8.15 Relation to Σ_1

Σ_1 studies ontic manifold scaling and approximate projectability in the large- N regime.

Σ_0 provides the conceptual basis for that paper. It interprets approximate projectability as the mathematical expression of physical isolation:

$$\text{physical isolation} \leftrightarrow \text{approximate projectability}.$$

Thus, Σ_1 should be read as developing one of the central technical conditions required by the ontology stated in Σ_0 .

Σ_0 says why the condition matters. Σ_1 studies how it can be controlled.

8.16 Summary

The role of Σ_0 is to sit beneath the technical corpus as an internal ontological foundation.

Its dependency position is:

$$\Sigma_0 \longrightarrow \Sigma_1 \longrightarrow \text{spacetime, locality, and large-}N \text{ bridge papers.}$$

Its interpretive position is:

$$\text{ontic foundation} \longrightarrow \text{finite-dimensional reconstruction} \longrightarrow \text{operational quantum predictions.}$$

It does not prove the existing technical results. It clarifies what they are about.

9 Research Targets

9.1 Purpose of This Section

The role of this section is to separate foundational commitments from future derivations. Σ_0 states the ontic picture required by CSD, but it does not complete the deeper derivation of that picture.

The programme therefore has two layers:

current ontological scaffold

and

future first-principles derivation.

The following research targets identify the main gaps between those two layers.

9.2 Deriving the Ontic Manifold Σ

The first open target is the derivation of Σ itself.

In the present paper, Σ is assumed to be a compact finite-dimensional Kähler manifold:

$$(\Sigma, \omega_\Sigma, J_\Sigma, g_\Sigma).$$

This assumption is structurally useful because it supports Hamiltonian dynamics, Liouville measure, and projection to projective quantum geometry. However, usefulness is not derivation.

A stronger theory should explain why physical reality should be represented by such a manifold. Possible derivation routes include:

1. constrained Hamiltonian geometry;
2. symplectic reduction;
3. finite-resolution physical state spaces;
4. information-geometric reconstruction;
5. operational indistinguishability classes;
6. stability requirements for deterministic measure-preserving dynamics.

The desired result would be a theorem or construction showing why a compact Kähler ontic manifold is required, or at least strongly selected, by deeper physical principles.

9.3 Deriving the Projection π

The second open target is the derivation of the projection

$$\pi : \Sigma \rightarrow \mathbb{CP}^{N-1}.$$

In the present paper, π is primitive structural data. It is smooth, surjective, and many-to-one, but its origin is not derived.

A deeper account should show why the epistemic quotient of Σ has the structure of complex projective space:

$$\mathbb{CP}^{N-1}.$$

Ideally, one would obtain

$$\mathbb{CP}^{N-1} \cong \Sigma / \sim,$$

where $x \sim y$ means that x and y are operationally indistinguishable in the relevant quantum-effective regime.

Such a derivation would need to explain:

1. why rays, rather than vectors, are the epistemic states;
2. why the quotient has complex projective geometry;
3. why the Fubini-Study metric and symplectic form appear;
4. why the relevant symmetry is $SU(N)$;
5. why fibres represent inaccessible ontic degrees of freedom.

Until this is achieved, π remains a controlled posit.

9.4 Deriving Approximate Projectability

The third open target is the deeper derivation of approximate projectability.

The present paper interprets approximate projectability as physical isolation:

$$\text{physical isolation} \leftrightarrow \text{approximate projectability}.$$

Mathematically, this means that fibre-sensitive terms remain below operational resolution:

$$\|\delta H_{\text{fb}}\|_{\mathcal{R},T} \leq \varepsilon.$$

Σ_1 develops this as a controlled regime condition. The deeper problem is to explain why ordinary isolated quantum systems should naturally satisfy it.

The target is a physical derivation of the suppression of fibre-sensitive dynamics from:

1. environmental decoupling;
2. timescale separation;
3. stability of projected dynamics;
4. symmetry constraints;
5. coarse-grained insensitivity to hidden fibre variation.

This would turn approximate projectability from a regime condition into an explained dynamical phenomenon.

9.5 Classifying Measurement Partitions

The fourth open target is the classification of physically realised measurement partitions.

The present paper assumes that a measurement context M determines outcome sectors

$$\Omega_i^\Sigma(M) \subseteq \Sigma_{\text{SA}},$$

or their epistemic images

$$\Omega_i^{\text{ep}}(M) \subseteq \mathbb{CP}^{N_{\text{SA}}-1}.$$

However, not every mathematical partition is physically admissible. A future theory should classify which partitions can be dynamically generated by realistic system-apparatus Hamiltonians.

The target is a result of the form:

$$H_{SA}, \text{ pointer stability, null-boundary conditions} \implies \{\Omega_i^\Sigma(M)\}.$$

Such a result would strengthen the measurement layer by replacing abstract admissibility with dynamical generation.

9.6 Deriving Composite Epistemic Structure

The fifth open target is the derivation of composite quantum state space from ontic principles.

For two subsystems, standard quantum mechanics uses

$$\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B,$$

and therefore

$$\mathbb{CP}^{N_A N_B - 1}$$

as the composite pure-state space.

CSD currently treats the relevant composite projective target as part of the effective finite-dimensional architecture. The deeper problem is to derive this structure from the joint ontic manifold:

$$\Sigma_{AB}.$$

The target is to explain how effective tensor-product kinematics emerges from non-factorising ontic structure:

$$\Sigma_{AB} \not\cong \Sigma_A \times \Sigma_B,$$

while still recovering subsystem marginals, entanglement, and no-signalling.

This is one of the main prerequisites for moving beyond finite-dimensional reconstruction toward a deeper account of many-body quantum structure.

9.7 Constructing the Spacetime Coarse-Graining Map

The sixth open target is the construction of a spacetime coarse-graining map.

The present paper distinguishes the epistemic projection

$$\pi : \Sigma \rightarrow \mathbb{CP}^{N-1}$$

from a future spacetime map

$$C_{st} : \Sigma \rightarrow M_{st}.$$

No explicit C_{st} is supplied here.

A future spacetime-facing paper must define how spacetime events, coordinates, metric distance, and causal order arise from Σ -level structure. The target is a construction of effective spacetime variables

$$X^\mu = X^\mu(x)$$

or spacetime events as equivalence classes of ontic data.

Without such a map, CSD remains primarily a quantum-foundational and finite-dimensional reconstruction programme, not yet a complete spacetime theory.

9.8 Recovering Lorentzian Metric Structure

The seventh open target is the recovery of Lorentzian geometry.

If spacetime is emergent, then the Lorentzian metric

$$g_{\mu\nu}$$

must be derived or recovered as an effective structure.

The desired result is not merely a map to a set of spacetime labels. It is a derivation of causal cones, invariant intervals, and relativistic propagation constraints from deeper Σ -level data.

A successful derivation would need to explain why the effective spacetime description has signature

$$(-, +, +, +),$$

or an equivalent convention, and why low-energy dynamics respect Lorentz symmetry to observed accuracy.

9.9 Relating c_Σ to the Observed Speed of Light

The eighth open target is the relationship between the Σ -level causal rate

$$c_\Sigma$$

and the observed speed of light

$$c.$$

The current paper does not identify these quantities. It only allows the possibility that c is the spacetime image of a deeper Σ -level causal bound.

The target is a derivation of one of the following forms:

$$c_\Sigma = c,$$

or

$$c = C_c(c_\Sigma, \mathcal{R}),$$

where C_c is a well-defined emergent-regime map.

This would connect ontic causal structure to relativistic causal structure.

9.10 Explaining Bell Correlations at the Σ Level

The ninth open target is the deep structural explanation of Bell correlations.

CSD already has a finite-dimensional Bell-facing layer. It can represent singlet correlations, no-signalling marginals, and context-indexed outcome sectors. However, this does not yet explain why spacetime-distant events should be structurally related at the Σ -level.

The proposed direction is stretched coarse-graining:

$$D_\Sigma(x, y) \text{ small}$$

while

$$D_{\text{st}}(C_{\text{st}}(x), C_{\text{st}}(y)) \text{ large.}$$

A future theory should show how such a relation can occur without operational signalling.

The target is therefore a theorem or model connecting:

1. joint ontic structure;
2. context-indexed outcome sectors;
3. spacetime-separated detector records;
4. no-signalling marginal stability;
5. Bell-violating joint correlations.

9.11 Clarifying Planck and Cosmological-Scale Bounds

The tenth open target is the status of candidate spacetime resolution bounds.

The present paper records the possible window

$$\ell_P \lesssim \Delta x_{\text{st}} \lesssim L_\Lambda,$$

but does not derive it.

A future theory must decide whether these scales are:

1. derived from Σ -level geometry;
2. inherited from empirical physics;
3. effective coarse-graining limits;
4. heuristic placeholders;
5. not fundamental to CSD at all.

Until then, the Planck and cosmological-constant scales should be treated as candidate guideposts, not as CSD predictions.

9.12 Connecting to Field-Like and Many-Mode Limits

The eleventh open target is the bridge from finite-dimensional models to field-like effective descriptions.

CSD currently works most cleanly at finite N . Many physical systems, however, are modelled using fields, local degrees of freedom, and continuum limits.

The target is to explain how field-like behaviour emerges from sequences of finite-dimensional models:

$$(\Sigma_N, \pi_N) \quad \text{as } N \rightarrow \infty,$$

without treating an infinite-dimensional ontic manifold as primitive.

This requires controlled convergence, effective locality, stable coarse variables, and many-mode coupling structure.

9.13 Priority Order

The immediate priority is not to attempt all research targets at once. The natural order is:

1. clarify approximate projectability through Σ_1 ;
2. construct the first spacetime coarse-graining proposal;
3. develop effective locality in large- N or many-mode regimes;
4. classify dynamically realised measurement partitions;
5. derive or constrain composite epistemic structure;
6. revisit the deeper derivation of Σ and π .

This order reflects dependency. Spacetime, locality, and Bell explanations cannot be made fully rigorous until approximate projectability and coarse-graining are under better control.

9.14 Summary

The present status of Σ_0 is:

ontology stated, derivation incomplete.

The paper's value lies in making the incomplete derivations explicit. The principal research targets are the derivation of Σ , the derivation of π , the explanation of approximate projectability, the classification of measurement partitions, the construction of emergent spacetime, the recovery of Lorentzian causal structure, and the explanation of Bell correlations at the Σ -level.

These targets define the next stage of the Σ -series.

10 Scope and Limitations

10.1 Purpose of the Scope Statement

The purpose of this section is to prevent overstatement. Σ_0 is an internal ontological foundation paper. It defines the commitments needed by the CSD programme, but it does not complete the programme.

The paper should be read as:

an assumptions audit + an ontological clarification + a roadmap for the Σ -series.

It should not be read as a completed first-principles derivation of quantum mechanics, spacetime, or relativistic physics.

10.2 What This Paper Establishes

This paper establishes the internal ontological reading of the CSD framework.

It states that the fundamental physical state is an ontic microstate

$$x \in \Sigma,$$

where Σ is a compact finite-dimensional Kähler manifold equipped with Liouville measure μ_L .

It states that quantum state space is epistemic:

$$[\psi] \in \mathbb{CP}^{N-1},$$

and that the relation between the ontic and epistemic levels is given by a smooth many-to-one projection

$$\pi : \Sigma \rightarrow \mathbb{CP}^{N-1}.$$

It states that measurement is a de-isolation event, in which a previously approximately projectable subsystem becomes dynamically coupled to apparatus or environmental degrees of freedom.

It states that CSD is deterministic, contextual, nonlocal relative to emergent spacetime, and constrained by operational no-signalling.

It states that spacetime is not treated as primitive in this paper, but as a future coarse-grained image of Σ -level structure.

10.3 What This Paper Does Not Establish

This paper does not derive Σ .

It does not derive the Kähler structure

$$(\omega_\Sigma, J_\Sigma, g_\Sigma).$$

It does not derive the projection

$$\pi : \Sigma \rightarrow \mathbb{CP}^{N-1}.$$

It does not derive the Fubini-Study geometry of projective quantum state space.

It does not derive approximate projectability from deeper dynamics.

It does not classify all physically admissible measurement partitions.

It does not derive composite tensor-product structure from ontic principles.

It does not construct the spacetime coarse-graining map

$$C_{\text{st}} : \Sigma \rightarrow M_{\text{st}}.$$

It does not derive Lorentzian metric structure, relativistic causal cones, or the observed speed of light.

It does not derive Planck-scale or cosmological-scale resolution bounds.

It does not provide a complete Σ -level mechanism for Bell correlations.

10.4 Status of the Main Objects

The current status of the central objects is as follows:

Σ is posited as the ontic manifold.

μ_L is the invariant Liouville measure induced by the symplectic form.

π is primitive structural data at this stage.

$\mathbb{CP}^{N-1}$ is the epistemic quantum state space.

C_{st} is not yet constructed.

c_Σ is a proposed Σ -level causal rate, not yet identified with c .

This status table is important because it distinguishes assumed structure from derived structure.

10.5 Finite-Dimensional Restriction

The present framework is finite-dimensional. The ontic manifold is assumed to satisfy

$$\dim_{\mathbb{R}} \Sigma < \infty.$$

This restriction is intentional. The existing CSD programme is strongest in finite-dimensional quantum reconstruction, finite N truncations, finite-dimensional measurement theory, and controlled continuum-like comparison.

The paper does not assume that an infinite-dimensional ontic manifold is physically fundamental. Field-like and continuum behaviour must instead be recovered through effective limits, nested truncations, or coarse-grained many-mode structures.

10.6 Spacetime Limitation

The paper does not provide a theory of spacetime emergence.

It introduces the need for a future map

$$C_{\text{st}} : \Sigma \rightarrow M_{\text{st}},$$

but does not define it.

Consequently, all statements about spacetime locality, spacetime distance, causal cones, and relativistic structure remain provisional. They are not yet consequences of the formalism.

The correct claim is:

CSD requires a spacetime emergence layer.

The incorrect claim would be:

CSD has already derived spacetime.

10.7 Bell-Sector Limitation

The Bell-facing position of Σ_0 is also limited.

The paper states that CSD is not Bell-local in the standard spacetime sense. It also states that CSD must preserve operational no-signalling.

However, it does not derive a full Σ -level structural mechanism for Bell correlations. It only identifies a possible route: spacetime-distant events may correspond to structurally close or jointly constrained regions in Σ .

The current claim is therefore:

Bell compatibility is supported by contextual outcome structure.

The stronger future claim would be:

Bell correlations are derived from Σ -level locality or adjacency.

That stronger claim remains open.

10.8 Measurement Limitation

The paper interprets measurement as de-isolation, but it does not classify all measurement interactions.

A complete measurement theory would need to show how physically realistic Hamiltonians generate stable outcome sectors:

$$H_{\text{SA}} \implies \{\Omega_i^{\Sigma}(M)\}.$$

The present paper assumes that such sectors exist for the measurement contexts already treated in the finite-dimensional operational framework. It does not prove a general classification theorem.

Thus, measurement is ontologically clarified, but not yet dynamically classified in full generality.

10.9 Interpretive Limitation

Σ_0 is not intended to persuade an external reader that CSD is already complete. It is intended to discipline the programme internally.

Its function is to say:

this is what the framework assumes,

this is what existing papers support,

and

this is what remains open.

That distinction is part of the scientific value of the paper.

10.10 Summary

The scope of Σ_0 is deliberately narrow.

It establishes the internal ontological foundation of CSD:

$$x \in \Sigma \longrightarrow [\psi] \in \mathbb{CP}^{N-1} \longrightarrow \text{operational quantum statistics.}$$

It does not complete the derivation of the framework.

The main limitation is therefore also the main contribution: Σ_0 makes the foundational assumptions explicit enough that later papers can attempt to derive, weaken, replace, or test them.

11 Conclusion

Σ_0 has stated the ontic commitments underlying Constraint-Surface Dynamics. Its central claim is that the fundamental physical state is not a Hilbert-space vector, a projective ray, or a density operator, but an ontic microstate

$$x \in \Sigma$$

evolving deterministically on a compact finite-dimensional Kähler manifold.

The projective quantum state

$$[\psi] \in \mathbb{CP}^{N-1}$$

is interpreted as an epistemic image of this deeper state under a smooth many-to-one projection

$$\pi : \Sigma \rightarrow \mathbb{CP}^{N-1}.$$

The projection identifies ontic distinctions that are not resolved at the operational quantum level.

The paper has also clarified the status of measurement. Measurement is not treated as primitive stochastic collapse. It is interpreted as a Σ -level de-isolation event, in which an approximately projectable subsystem becomes dynamically coupled to apparatus or environmental degrees of freedom. Definite outcomes arise through deterministic evolution into context-defined outcome sectors.

The Bell-facing position is likewise explicit. CSD is deterministic and contextual. It is nonlocal relative to emergent spacetime, but constrained by operational no-signalling. It does not assume Bell-local factorisation or a global assignment of outcomes for all incompatible measurement contexts.

The paper has deliberately not claimed more than this. It has not derived Σ , π , spacetime, Lorentzian metric structure, c , Planck-scale bounds, cosmological-scale bounds, or the full Σ -level mechanism behind Bell correlations. These remain research targets.

The contribution of Σ_0 is therefore foundational rather than demonstrative. It makes explicit the ontology that the existing CSD technical corpus already uses. It separates posits from derived results. It identifies the open derivations required for later work.

In this role, Σ_0 provides the internal base for the Σ -series:

$$\Sigma_0 \longrightarrow \Sigma_1 \longrightarrow \text{emergent spacetime, causal order, effective locality, and large-}N \text{ structure.}$$

The immediate next step is to use Σ_1 to sharpen approximate projectability as the controlled mathematical expression of physical isolation. Later papers must then address the emergence of spacetime, the recovery of Lorentzian causal structure, and the relation between Σ -level geometry and spacetime-local phenomenology.

Thus, Σ_0 does not complete CSD. It makes the programme harder to misstate, easier to audit, and better prepared for the next technical stage.

A Glossary of Σ -Level and Spacetime-Level Terms

A.1 Purpose of the Glossary

This appendix records the main terminology used in Σ_0 . Its purpose is to keep the distinction between ontic, epistemic, operational, and spacetime-level language explicit.

The central distinction is:

$$\text{ontic structure} \neq \text{epistemic quantum representation} \neq \text{spacetime operational record.}$$

A.2 Core Terms

Term	Σ -level meaning	Epistemic or spacetime image
Ontic state	Actual microstate $x \in \Sigma$	Not directly observed
Quantum state	Not fundamental	Projected ray $[\psi] \in \mathbb{CP}^{N-1}$
Projection	Map $\pi : \Sigma \rightarrow \mathbb{CP}^{N-1}$	Quantum-level representation
Fibre	$F_{[\psi]} = \pi^{-1}([\psi])$	Operationally unresolved ontic variation
Liouville measure	μ_L on Σ	Source of volume weights
Fubini-Study measure	Not ontic primitive	Canonical projective measure μ_{FS}
Hamiltonian	$H_\Sigma : \Sigma \rightarrow \mathbb{R}$	Operator-induced energy in quantum-effective regime
Time	t_Σ	Laboratory time t after coarse-graining
Distance	$D_\Sigma(x, y)$	Spacetime distance after coarse-graining
Causal rate	c_Σ	Candidate source of observed c
Measurement	De-isolation event	Detector record or pointer state
Outcome sector	$\Omega_i^\Sigma(M)$	$\Omega_i^{\text{ep}}(M)$ or operational outcome i
Probability	Normalised volume weight	Born-form operational weight

Table 1: Core distinction between Σ -level ontology and its epistemic or spacetime images.

A.3 Ontic Layer

The ontic layer is the level of actual physical state and deterministic dynamics.

The basic ontic state is

$$x \in \Sigma.$$

The ontic trajectory is

$$x(t_\Sigma) = \Phi_{t_\Sigma}^\Sigma(x_0).$$

The ontic Hamiltonian is

$$H_\Sigma : \Sigma \rightarrow \mathbb{R}.$$

The invariant measure is

$$\mu_L.$$

This layer is not directly observed. It is inferred from the role it plays in explaining projected quantum statistics and measurement structure.

A.4 Epistemic Layer

The epistemic layer is the quantum state-space representation obtained after projection:

$$\pi : \Sigma \rightarrow \mathbb{CP}^{N-1}.$$

The epistemic state is

$$[\psi] \in \mathbb{CP}^{N-1}.$$

The projective state does not identify a unique ontic microstate. Instead, it identifies a fibre:

$$F_{[\psi]} = \pi^{-1}([\psi]).$$

The epistemic layer is therefore a compressed description of the ontic layer.

A.5 Operational Layer

The operational layer is the level of laboratory predictions and measurement records.

For projective measurements, probabilities take the form

$$p_i = \langle \psi | \hat{P}_i | \psi \rangle.$$

For generalised measurements, they take the form

$$p_i = \text{Tr}(\rho \hat{E}_i).$$

In Σ_0 , these are not primitive stochastic laws. They are operational representations of projected volume structure under the finite-dimensional CSD dictionary.

A.6 Spacetime Layer

The spacetime layer is the level of effective localisation, laboratory time, detector separation, and causal structure.

A future spacetime coarse-graining map is denoted schematically by

$$C_{\text{st}} : \Sigma \rightarrow M_{\text{st}}.$$

This map is not the same as the epistemic projection π . The distinction is:

$$\pi : \Sigma \rightarrow \mathbb{CP}^{N-1} \quad \text{projects to quantum state space,}$$

whereas

$$C_{\text{st}} : \Sigma \rightarrow M_{\text{st}} \quad \text{coarse-grains to spacetime structure.}$$

The construction of C_{st} is not supplied in Σ_0 .

A.7 Measurement Terms

A measurement context is denoted by

$$M.$$

The corresponding ontic outcome regions are

$$\Omega_i^\Sigma(M) \subseteq \Sigma_{\text{SA}}.$$

The corresponding epistemic outcome regions are

$$\Omega_i^{\text{ep}}(M) \subseteq \mathbb{CP}^{N_{\text{SA}}-1}.$$

A realised outcome occurs when

$$x_{\text{SA}}(T) \in \Omega_i^\Sigma(M).$$

Measurement is therefore not defined as collapse. It is defined as deterministic sector realisation following system-apparatus coupling.

A.8 Recommended Usage

Use $x \in \Sigma$ for ontic microstates.

Use $[\psi] \in \mathbb{CP}^{N-1}$ for epistemic pure states.

Use ρ for operational mixed-state representation.

Use μ_L for Liouville measure on Σ .

Use μ_{FS} for Fubini-Study measure on projective space.

Use Ω_i^Σ for ontic outcome sectors.

Use Ω_i^{ep} for epistemic outcome regions.

Use t_Σ for the ontic evolution parameter.

Use t only for emergent or laboratory time.

Use C_{st} for the future spacetime coarse-graining map.

A.9 Terms to Avoid or Use Cautiously

The word “collapse” should be avoided unless discussing standard quantum mechanics. In CSD, use “sector realisation”, “outcome-conditioned update”, or “operational conditionalisation”.

The phrase “hidden variable” should be used cautiously. The ontic state $x \in \Sigma$ is not a Bell-local hidden variable in the standard factorisation sense.

The phrase “local” should specify the relevant level. CSD is not Bell-local in emergent spacetime. It may seek a deeper form of Σ -level locality, but that has not yet been derived.

The word “derive” should be used only where the relevant result is actually established. For Σ , π , spacetime, and c_Σ , use “posit”, “assume”, “introduce”, or “identify as a research target” unless a later paper supplies the derivation.

B Comparison with Neighbouring Programmes

B.1 Purpose of the Comparison

This appendix situates Σ_0 relative to neighbouring approaches in quantum foundations and emergent spacetime. The aim is not to provide a full literature review. The aim is to clarify what CSD is, what it is not, and where its ontological commitments differ from better-known frameworks.

The central distinction is:

CSD is not merely a reformulation of quantum mechanics on projective space.

It introduces a deeper ontic manifold

$$\Sigma$$

and treats projective quantum state space as an epistemic image:

$$\Sigma \xrightarrow{\pi} \mathbb{CP}^{N-1}.$$

This places CSD near geometric quantum mechanics, but not inside it. It also makes CSD structurally different from Bohmian mechanics, Everettian quantum mechanics, collapse theories, and purely epistemic interpretations.

B.2 Geometric Quantum Mechanics

Geometric quantum mechanics reformulates finite-dimensional quantum theory on projective Hilbert space. Pure states are rays

$$[\psi] \in \mathbb{CP}^{N-1},$$

and $\mathbb{CP}^{N-1}$ carries the Fubini-Study Kähler structure

$$(\omega_{FS}, J_{FS}, g_{FS}).$$

In this setting, Schrödinger evolution can be represented as Hamiltonian flow on projective state space, and expectation-value functions correspond to geometric observables.

CSD accepts this geometric structure but changes its ontological status. In CSD, $\mathbb{CP}^{N-1}$ is not the fundamental physical state space. It is the epistemic image of a deeper ontic manifold:

$$\pi : \Sigma \rightarrow \mathbb{CP}^{N-1}.$$

Thus, geometric quantum mechanics supplies the projective geometry. CSD adds an ontic layer beneath it.

The relationship is:

$$\text{geometric quantum mechanics} = \text{geometry of the epistemic layer},$$

whereas

$$\text{CSD} = \text{ontic geometry plus epistemic projection}.$$

B.3 Bohmian Mechanics

Bohmian mechanics is deterministic and realist. It introduces actual particle configurations guided by a wavefunction. In its standard nonrelativistic form, the ontology includes particle positions in configuration space, while the wavefunction evolves by the Schrödinger equation and guides particle motion.

CSD is also deterministic and realist, but its ontology is different.

In CSD, the fundamental state is

$$x \in \Sigma.$$

The wavefunction is not a guiding physical field. It is an epistemic representative of the projected ray

$$[\psi] = \pi(x).$$

Thus, CSD does not add particle trajectories guided by ψ . It replaces the primitive wavefunction ontology with deterministic dynamics on Σ .

Both CSD and Bohmian mechanics are nonlocal relative to ordinary spacetime if they reproduce Bell-violating correlations. The difference lies in the structure of the ontology:

Bohmian mechanics: particles plus guiding wave,

CSD: ontic Kähler manifold plus projection.

B.4 Everettian Quantum Mechanics

Everettian quantum mechanics retains universal unitary evolution and interprets branching structure as physically real. The universal wavefunction is normally treated as the central ontological object.

CSD does not take this route.

CSD is single-trajectory:

$$x(t_\Sigma) \in \Sigma.$$

There is one realised ontic history. Measurement outcomes are not many equally realised branches. They are context-defined sector realisations along the actual ontic trajectory.

The operational appearance of branching may still occur in the projected description, especially during measurement and decoherence-like processes. However, CSD does not interpret all projected branches as separate realised worlds.

The contrast is therefore:

Everett: universal wavefunction and branching worlds,

CSD: single ontic trajectory and context-defined outcome sectors.

B.5 Collapse Theories

Collapse theories modify standard quantum dynamics by adding stochastic collapse processes. The purpose is to produce definite outcomes dynamically rather than by measurement postulate.

CSD also aims to account for definite outcomes, but it does not introduce stochastic collapse.

At the ontic level, the trajectory remains deterministic:

$$x(t_\Sigma) = \Phi_{t_\Sigma}^\Sigma(x_0).$$

The apparent update after measurement is interpreted as outcome-conditioned restriction to the realised sector:

$$x_{SA}(T) \in \Omega_i^\Sigma(M).$$

Thus, CSD does not alter Schrödinger dynamics by a random collapse law. It treats collapse-like update as an operational representation of deterministic sector realisation.

The contrast is:

collapse theories: stochastic physical collapse,
CSD: deterministic de-isolation and conditional update.

B.6 QBism and Epistemic Interpretations

QBism and related epistemic approaches treat the quantum state as a representation of information, belief, expectation, or agent-centred probability assignment.

CSD agrees with epistemic interpretations on one point:

ψ is not the fundamental ontic state.

However, CSD does not reduce the theory to agent-centred probabilities. It posits an observer-independent ontic substrate:

$$x \in \Sigma.$$

The epistemic state

$$[\psi] \in \mathbb{CP}^{N-1}$$

is a projected representation of deeper physical structure, not merely a personal betting commitment.

Thus, CSD combines an epistemic reading of the quantum state with a realist ontology beneath it.

B.7 Classical Statistical Mechanics

CSD has a structural analogy with classical statistical mechanics. In both cases, probabilities can arise from measures over underlying state spaces rather than from primitive stochastic laws.

In classical statistical mechanics, one often uses phase space, Liouville measure, and coarse-grained macrostates. In CSD, one uses

$$\Sigma, \quad \mu_L, \quad \Omega_i^\Sigma(M).$$

The analogy is useful but limited. CSD is not classical statistical mechanics applied directly to particles in spacetime. Its state space is a compact Kähler ontic manifold projected to quantum state space. Its measurement sectors are contextual and must reproduce quantum constraints.

The analogy is therefore:

volume measure and coarse-graining
rather than
ordinary classical phase-space mechanics.

B.8 Hidden-Variable Theories

CSD can be described broadly as an ontological completion, but it should not be described as a Bell-local hidden-variable theory.

The ontic state is

$$x \in \Sigma,$$

but CSD does not assume Bell-local factorisation:

$$P(s, t|a, b, \lambda) = P_A(s|a, \lambda)P_B(t|b, \lambda).$$

Nor does it assume global preassigned values for all incompatible measurement settings.

Instead, outcome maps are context-indexed:

$$R_{a,b} : \Sigma_{\text{SA}} \rightarrow \{+1, -1\} \times \{+1, -1\}.$$

This makes CSD deterministic and contextual, not Bell-local.

The phrase “hidden variable” is therefore potentially misleading unless qualified. CSD contains ontic structure beyond the quantum state, but it does not preserve the assumptions ruled out by Bell-type arguments.

B.9 Causal-Set and Emergent-Spacetime Programmes

Causal-set theory treats causal order as fundamental and spacetime geometry as emergent from a discrete partial order.

CSD does not currently posit a causal set. Its primitive structure is not a discrete causal order but a compact Kähler ontic manifold:

$$(\Sigma, \omega_\Sigma, J_\Sigma, g_\Sigma).$$

However, CSD shares with emergent-spacetime programmes the idea that spacetime need not be fundamental. In Σ_0 , spacetime is treated as a future coarse-grained image:

$$C_{\text{st}} : \Sigma \rightarrow M_{\text{st}}.$$

The difference is that CSD begins from quantum-foundational geometry and works outward toward spacetime, rather than beginning from causal order and reconstructing geometry.

B.10 Holography and ER-EPR

Holographic approaches suggest that spacetime geometry may be deeply related to quantum information, entanglement, and boundary descriptions. ER-EPR proposals suggest a possible relation between entanglement and spacetime connectivity.

CSD does not derive holography. It does not derive wormhole geometry. It does not currently claim that entanglement is equivalent to geometric connection in spacetime.

However, CSD is compatible with the broad idea that spacetime relations may emerge from deeper non-spacetime structure. In particular, the stretched coarse-graining idea in Σ_0 allows the possibility that states close in Σ may appear distant in spacetime:

$$D_\Sigma(x, y) \text{ small,}$$

while

$$D_{\text{st}}(C_{\text{st}}(x), C_{\text{st}}(y)) \text{ large.}$$

This resemblance is conceptual only. It is not yet a technical connection to holography or ER-EPR.

B.11 Superdeterminism

CSD should not be casually identified with superdeterminism.

Superdeterministic models typically evade Bell constraints by denying statistical independence between hidden variables and later measurement settings. CSD’s Bell-facing strategy is different. It rejects Bell-local factorisation and global context-independent outcome assignment, while preserving operational no-signalling.

The relevant structure is contextuality:

$$\Omega_{st}^{\Sigma}(a, b),$$

not necessarily a conspiracy between preparation variables and future setting choices.

That said, a complete CSD account of measurement settings must eventually specify how apparatus choices are represented in Σ . Until that is done, the relationship to statistical independence assumptions should be stated carefully.

The safe statement is:

CSD is contextual and non-Bell-local, not standard superdeterminism.

B.12 Relational Quantum Mechanics

Relational quantum mechanics treats quantum states as relative to physical systems rather than as absolute observer-independent states.

CSD agrees that the quantum state is not an absolute ontic object. However, it does not make relational state assignment fundamental. It posits an underlying ontic state

$$x \in \Sigma$$

and interprets quantum states as projected descriptions.

Thus, CSD differs from relational approaches by retaining a single underlying ontic state, even though operational descriptions may be relational, contextual, or coarse-grained.

B.13 Summary

CSD is closest to geometric quantum mechanics in mathematical language, but differs by adding an ontic manifold beneath projective quantum state space:

$$\Sigma \xrightarrow{\pi} \mathbb{CP}^{N-1}.$$

It is realist and deterministic like Bohmian mechanics, but without a guiding wavefunction ontology.

It is single-trajectory rather than Everettian.

It explains update without stochastic collapse.

It treats the quantum state as epistemic, but not merely subjective.

It is contextual and nonlocal relative to spacetime, but constrained by no-signalling.

Its distinctive claim is therefore the combination:

compact Kähler ontology+many-to-one epistemic projection+Liouville volume typicality+context-indexed measurement sectors+em

That combination is the specific ontological profile of Σ_0 .

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